

Asha Shah

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Summary

Senior Quality Assurance Engineer with several years leading automation-first testing for cloud-native, high-availability platforms. Architected and maintained 80+ automated test suites in Playwright and Selenium, cutting regression cycles by 35% and boosting release velocity by 40% across three development teams. Established end-to-end test governance and compliance controls that achieved zero audit findings, directly aligning with the rigorous regulatory standards of insurance and financial services. Eager to apply this expertise to drive quality, risk mitigation, and faster delivery of mission-critical financial products.

Skills

- **Testing & QA:** Playwright, Cypress, Selenium, pytest, JUnit, TestNG, manual testing, regression testing, smoke testing, exploratory testing, ETL testing, performance testing, API contract testing, test case design, boundary value analysis, equivalence partitioning
- **Test Management & Reporting:** JIRA, Linear, Azure DevOps, TestRail, Zephyr, Allure Reports, test plan authoring, traceability matrices, defect lifecycle management, compliance audit documentation
- **Automation Frameworks & Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash/Shell, Page Object Model (POM), data-driven testing, keyword-driven testing, BDD (Cucumber, Gherkin)
- **DevOps & CI/CD:** Jenkins, GitHub Actions, Docker, AWS Fargate, Azure DevOps, Git, GitLab, quality gates, pipeline integration, shift-left testing
- **Monitoring & Observability:** AWS CloudWatch, Azure Monitor, Grafana, Datadog, pipeline health tracking, SLA monitoring
- **Software Development & Architecture:** Java, Node.js, Spring Boot, TypeScript, REST, GraphQL, Microservices, Vue.js, Meteor, MongoDB
- **Leadership & Methodologies:** Agile/Scrum, cross-functional synergy, technical documentation, Agile Engineering, Kanban, XP

Work Experience

Southern Illinois University Edwardsville

Jun 2026 - Present

Volunteer Researcher – Artificial Intelligence & NLP

- Conduct AI and NLP research on semantic search, information retrieval, text embeddings, and clustering for over 1,000 low-resource Sanskrit and English texts, using Python and Hugging Face libraries to build and evaluate models, which expands the searchable corpus for multilingual research
- Design and validate experiments that compare embedding-based and traditional retrieval methods, using Python, MongoDB, and evaluation metrics such as MAP to identify accuracy improvements and edge-case behavior, resulting in a clear recommendation for the most effective approach
- Develop automated evaluation workflows in Python that test retrieval methods and verify experimental outputs, cutting manual testing time by half and ensuring reproducible results across runs
- Research AI-assisted and agent-based techniques for automating software testing, test generation, and quality analysis, prototyping scripts with Playwright and managing code with Git, which identified viable automation strategies for future implementation
- Develop an interactive research platform with React and JavaScript for semantic search and network visualization, integrating MongoDB for data storage and conducting functional validation of search and user workflows, which enables researchers to explore the text corpus more efficiently

Keela.Co

Jan 2021 - Jul 2024

Senior Quality Assurance Engineer

- Architected 80+ automated test suites using Playwright and Selenium in TypeScript, reducing regression cycles by 35% and boosting release velocity by 40% across three dev teams.
- Engineered end-to-end REST API test coverage for PayPal, Salesforce, and Twilio integrations using Postman and Python, ensuring 99.9% uptime across cloud-native SaaS on AWS and Google Cloud.
- Leveraged AI-enhanced testing workflows and GPT-based test case generation in Playwright, accelerating test authoring by 30% and improving edge case coverage across regression suites
- Designed reusable page object models and shared CI hooks in Playwright and TypeScript, standardizing test architecture and cutting framework onboarding time by 50% for junior engineers.
- Implemented quality gates in Jenkins and GitHub Actions pipelines, blocking deployments on critical test failures and eliminating manual release checks across three parallel dev teams.
- Authored 100+ compliance test cases and traceability matrices in JIRA, achieving zero audit failures across all regulatory review cycles for cloud-hosted SaaS features.
- Resolved 900+ defects using data-driven root cause analysis in JIRA, improving product stability by 40% and earning internal QA Innovator of the Year recognition.
- Mentored 5 junior QA engineers using AI-enhanced workflows and tailored training plans, cutting onboarding time by 50% and raising automation coverage across the team.
- Led test strategy and acceptance criteria definition for 5 enterprise-grade features in Agile/Scrum sprints, delivering all releases ahead of schedule with zero post-deploy critical defects.

Spiralogsics, Inc.

Oct 2020 - Jan 2021

DevSecOps Engineer Intern

- Developed a Java-based web platform with JWT authentication, encryption controls, and secure coding practices, enhancing user data protection
- Identified and remediated SQL injection vulnerabilities, strengthening application security posture and achieving zero security incidents post-deployment.
- Automated SSL certificate lifecycle management, eliminating manual intervention and improving operational reliability.
- Built and maintained Docker-based CI/CD pipelines deployed on AWS Fargate, reducing deployment times by 25% and supporting continuous delivery initiatives.
- Collaborated with development and infrastructure teams to embed DevSecOps security validation into deployment workflows, resulting in earlier detection of vulnerabilities and improved compliance

PROJECTS

EvidenceCV: Evidence-Based Resume Generation using RAG **Mar 2026**

- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline to ground resume content in verified evidence from GitHub, research papers, and certifications.
- Built a Python-based ingestion system to pull data from GitHub APIs and PDFs, generating vector embeddings with Sentence Transformers (all-MiniLM-L6-v2).
- Implemented semantic retrieval with HNSW indexing in ChromaDB for job-aware evidence matching between candidate skills and ATS-specific job requirements.
- Developed an automated LaTeX rendering service to generate professional, ATS-optimized resumes with consistent formatting and keyword alignment.
- Delivered a functional full-stack system (React/Vite frontend) within 36 hours, earning 3rd place at the eHacks Hackathon.

RikNetra: Multimodal Rigveda Network Visualization (D3.js, JavaScript, NLP) riknetra.net **Jan 2025 - Oct 2025**

- Developed a groundbreaking platform for exploring 1,000+ interconnected Rigveda hymns using real-time semantic search, AI-powered filtering, and weighted graph visualizations. Leveraged NLP and D3.js to deliver an intuitive interface, ideal for scalable data integration testing in academic and enterprise settings.

Blood Cancer Detection AI

- Developed a TensorFlow and Scikit-learn classification model achieving 92% accuracy in early blood cancer detection, applying cross-validation and confusion matrix analysis to ensure reliable diagnostic performance.
- Engineered Python-based data preprocessing pipelines handling medical imaging datasets, implementing feature normalization, train/test isolation, and class imbalance handling to prevent data leakage.
- Integrated model inference with cloud-based data pipelines for real-time diagnostic analysis, validating end-to-end prediction latency and output consistency across test environments.

Keela CRM Reimagination

- Re-engineered a cloud-native CRM with real-time data synchronization and zero-downtime authentication, optimizing performance for 10,000+ users. Implemented robust ETL processes and secure APIs, demonstrating proficiency in data-driven QA and microservices architecture.

EDUCATION

Southern Illinois University Edwardsville
Master of Science, Computer Science

Aug 2024 - May 2026

Certifications

- ISTQB® Certified Tester – Foundation Level (CTFL)
- Microsoft Azure DevOps Engineer Expert (AZ-400)